

ACKNOWLEDGEMENT

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We are also genuinely thankful to all the individuals who participated in our research, provided crucial insights, and helped us gather the necessary information to design a practical and user-focused system. Your input played a vital role in aligning the project with real-world needs.

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Once again, we sincerely thank each and every person who played a role in the realization of this project. Your contributions have helped make **e-Liquor** a successful and rewarding experience. We are truly grateful.

ABSTRACT

This report focuses on the development of **e-Liquor**, an e-commerce platform dedicated to the online sale and late-night delivery of liquor within the Kathmandu Valley. The primary objective of the project is to create a convenient, secure, and user-friendly system that caters to the growing demand for digital liquor delivery services in urban areas. The report outlines the project's objectives, limitations, and the chosen development methodology, which follows the waterfall model for its structured and phase-based approach.

The report explores the integration of key technologies and features essential for a reliable e-commerce system, with emphasis on secure user authentication, efficient order management, responsive design, and streamlined delivery tracking. Notable features include user registration, product browsing, cart functionality, order placement, and an admin panel for managing inventory and orders, all aimed at enhancing the customer experience.

In addition, the report details the system's analysis, design, implementation, and testing phases, ensuring the platform operates with accuracy and consistency. The conclusion highlights the platform's success in addressing the specific logistical and regulatory challenges of liquor delivery, while also suggesting future improvements such as digital payment integration, mobile app development, and enhanced personalization features.

Overall, **e-Liquor** is presented as a scalable and practical solution for online liquor sales, with promising opportunities for expansion and innovation in the digital retail market.

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CHAPTER 1: INTRODUCTION

1.1 Project Introduction

The project aims to introduce **e-Liquor**, an innovative e-commerce platform dedicated to the online sale and timely delivery of liquor products within Kathmandu Valley. As digital commerce continues to reshape consumer behavior, there is a growing demand for specialized platforms that provide a safe, convenient, and user-friendly liquor shopping experience. **e-Liquor** seeks to fulfill this demand by offering a dedicated online space where customers can browse, select, and order a wide range of alcoholic beverages from the comfort of their homes—especially during late-night hours.

The primary objective of **e-Liquor** is to establish a platform that caters specifically to individuals seeking quality liquor brands with fast and reliable delivery. By collaborating with licensed vendors and suppliers, **e-Liquor** aims to ensure authenticity, proper inventory management, and adherence to legal regulations regarding the sale and distribution of alcohol.

The platform will feature an intuitive and responsive web interface designed to make the shopping experience smooth and enjoyable. Users will be able to search and filter products by type, brand, or price range, and quickly locate their preferred items.

To maintain a secure and trustworthy environment, **e-Liquor** will integrate robust user authentication, age verification checks, and secure payment gateways. Real-time order tracking and efficient delivery services will ensure that customers are always informed about the status of their purchases, from checkout to doorstep delivery—especially during the platform’s core service hours of 7 PM to 2 AM.

By combining convenience, selection, and safety, **e-Liquor** aims to become the leading online destination for liquor purchases in Kathmandu Valley.

1.2 Problem of Statement

The traditional liquor retail industry is facing growing challenges with the increasing shift toward online shopping, especially in urban areas like Kathmandu Valley. Despite the rising demand for convenient, late-night access to alcoholic beverages, there is a noticeable lack of a specialized e-

commerce platform dedicated solely to liquor sales and delivery. This gap creates difficulties for both consumers and licensed liquor vendors.

For consumers, purchasing liquor through existing platforms is often inconvenient or not possible during late-night hours. Many general e-commerce sites do not offer a wide selection of alcoholic beverages, and even when available, they lack key features such as product categorization by type, real-time stock visibility, and efficient delivery tracking. Furthermore, the absence of proper age verification and secure payment options on many platforms adds to customer concerns regarding safety and legality.

In summary, the lack of a specialized and secure online liquor delivery platform creates a fragmented experience for customers and limits the digital reach and operational efficiency for businesses. There is a clear need for a well-structured, compliant, and user-focused system that bridges this gap in the local market.

1.3 Objectives of the Project

The following is an overview of the project's main goals:

- Develop a user-friendly platform.
- Create a diverse collection of lens and glasses.
- Provide accurate product information.
- Facilitate secure transactions.
- Enable social sharing and customer engagement.

1.4 Limitations of the Project

- Limited brand partnerships.
- Challenges in inventory management.
- Building customer trust and brand awareness.
- Difficulties in payment integration.
- Technical constraints.
- Logistics of shipping and delivery.

1.5 Assignment of roles and responsibilities

Team Members: Anish Nepal, Bigyan Khatri , Unik Oli

1. Anish Nepal – Backend & API Developer (Laravel + MySQL)

Responsibilities:

- Set up and maintain the Laravel backend structure
- Design and develop RESTful APIs for product listings, cart, user authentication, order processing, etc.
- Implement user roles (admin, customer) and authentication
- Integrate payment gateway (if applicable)
- Manage database schema and queries using MySQL
- Handle server-side validation, business logic, and security
- Provide API documentation for frontend integration

2. Bigyan Khatri – Frontend Developer (React + UI/UX Integration)

Responsibilities:

- Build user-facing interfaces using React
- Integrate with backend APIs for dynamic product display, user authentication, shopping cart, and orders
- Develop responsive layouts and reusable components
- Handle client-side validation and error handling
- Optimize performance and ensure cross-browser compatibility
- Assist with UI design using HTML/CSS/JavaScript as needed

3. Unik Oli – Full-stack Support (HTML/CSS/JS)

Responsibilities:

- Design wireframes and mockups for the website (homepage, product pages, checkout, etc)
- Implement clean and modern UI using HTML/CSS/JavaScript

- Ensure mobile responsiveness and accessibility standards
- Support backend and frontend members with layout, styling, and UI fixes
- Test full application flow and report bugs
- Coordinate team meetings, version control (Git), and deployment assistance

1.6 Report Organization

This report is structured into five well-defined chapters, each focusing on a critical stage in the development of the **e-Liquor** e-commerce platform. The organization of the content ensures a smooth and logical progression from the conceptualization of the idea to its practical execution and evaluation. A brief overview of each chapter is provided below:

Chapter 1: Introduction – Covers the project overview, problem statement, objectives, limitations, team roles, and report structure.

Chapter 2: Requirement and Feasibility Analysis – Includes analysis of existing systems, selected development model, system requirements, and feasibility study.

Chapter 3: System Design – Presents system architecture, database design, process flow diagrams, and any relevant algorithms.

Chapter 4: Implementation – Describes the development process, tools used, and testing phases including unit, integration, and system testing.

Chapter 5: Conclusion and Future Enhancement – Summarizes the project outcomes and suggests possible future improvements.

CHAPTER 2: REQUIREMENT AND FEASIBILITY ANALYSIS

2.1: Existing Systems Overview

System 1: Cheers Online Store Nepal

Cheers Online Store Nepal is a well-known online liquor delivery platform that operates within Kathmandu and other major cities in Nepal. It provides customers with a wide selection of alcoholic beverages, including beer, whiskey, wine, vodka, and rum. Cheers offers convenient features such as home delivery, product categorization, and online payment options.

Pros:

- Wide product range covering various types of liquor (beer, whiskey, rum, wine, etc.)
- User-friendly categorization and basic filtering options
- Home delivery service available across key urban areas
- Multiple payment methods including digital wallets and cash on delivery

Cons:

- Limited personalization features (no recommendations based on user behavior)
- High system load can occasionally lead to slow performance
- Age verification process is minimal and can be bypassed.

System 2: Daraz

Introduction: In recent years, Daraz has also made liquor products available through selected vendors, allowing users to purchase alcoholic beverages online with home delivery service in certain regions.

Pros:

- Simple and user-friendly interface
- Competitive pricing
- Frequent discounts and deals

- Easy order customization

Cons:

- Liquor availability is limited to certain sellers and regions.
- Delivery times may vary and are not optimized for urgent or late-night needs.
- Not a dedicated beverages supplier.

2.2: System development model

The Waterfall Model was selected for the e-liquor project due to its structured and sequential approach, which aligns well with the clearly defined project scope and requirements.

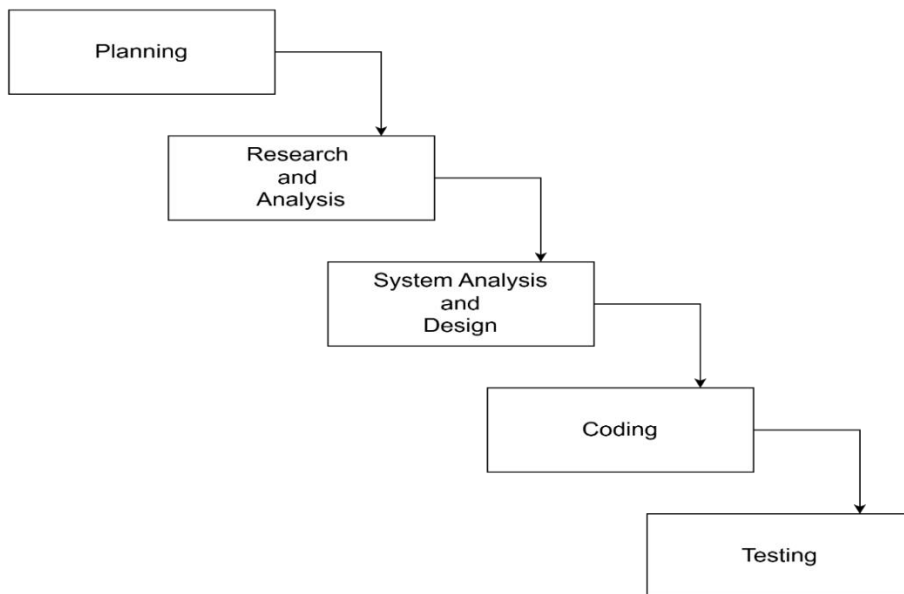


Figure 1 Waterfall model

- Well-defined Requirements: The project has clear and fixed requirements from the beginning, making the linear flow of the Waterfall model suitable.
- Simple and Structured Approach: Waterfall allows for disciplined development with clear milestones at each phase (analysis, design, implementation, testing, deployment, and maintenance).
- Ease of Management: It is easier to manage progress, deadlines, and responsibilities as each phase must be completed before moving to the next.

- Ideal for Academic Projects: For a short-term academic project like e-Liquor, the Waterfall model ensures that all documentation and phases are completed in an orderly and traceable way.

This model ensures smooth project execution and timely delivery, especially when there is limited risk of frequent changes in requirements.

2.3 Requirement Analysis

2.3.1 Functional Requirements

Essential features and capabilities define the functional requirements of a liquor-selling e-commerce website. These requirements shape the website's development, ensuring it meets customer expectations and delivers a seamless shopping experience. By understanding and addressing these requirements, the website can enhance usability, satisfaction, and business success.

Some functional requirements for this project are listed below:

- User Registration & Login (with CAPTCHA)
- Product catalog and filtering
- Order placement and payment via eSewa
- Admin panel for CRUD operations
- Order tracking and receipt generation

A use case diagram is a visual representation of how actors interact with a system. It shows the different actions users can perform and their relationships to the system. It helps with requirements gathering, system design, and stakeholder communication. Use case diagram for our project is shown below:

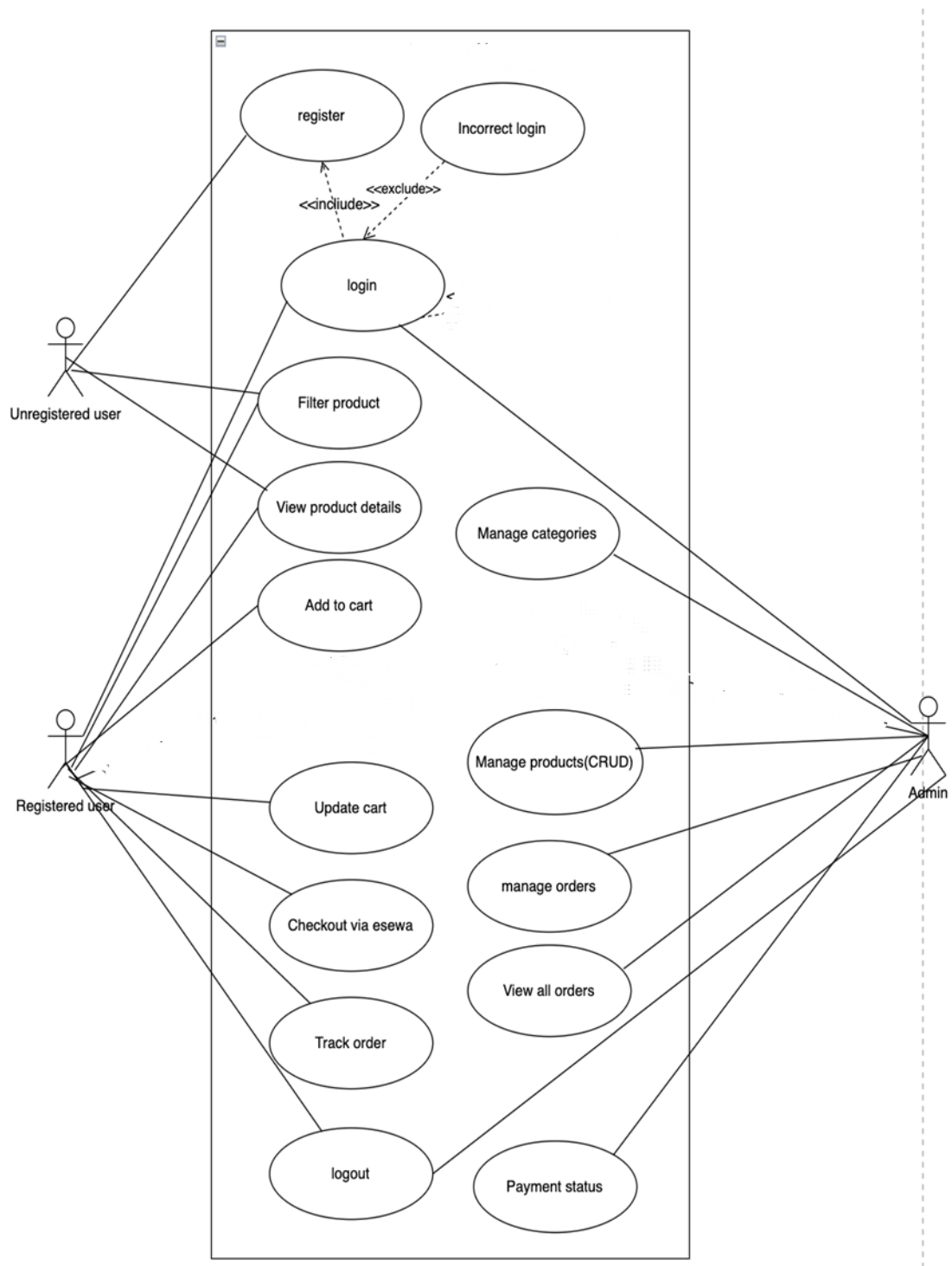


Figure 2 Use Case Diagram for e-Liquor

2.3.2 Non-Functional Requirements

- **Usability:** The system should have a user-friendly interface, with intuitive navigation and clear instructions, to ensure ease of use for both customers and administrators. It should provide a seamless and pleasant user experience.
- **Reliability & Availability:** The system should be highly reliable and available, ensuring minimal downtime and uninterrupted access to the e-commerce website. It should have robust backup and recovery mechanisms to prevent data loss and ensure system stability.
- **Performance:** The system should be able to handle a large number of concurrent users and process transactions efficiently. It should provide fast response times, quick page loading, and smooth browsing and checkout experiences to ensure optimal performance.
- **Security:** The system should implement strong security measures to protect customer data, including personal information, payment details, and order history. It should have secure authentication and authorization mechanisms, encryption protocols, and robust protection against common security threats, such as hacking and data breaches.
- **Scalability:** The system should be designed to handle future growth and accommodate increasing user traffic and data volume. It should be scalable and able to handle additional product listings, users, and transactions without significant performance degradation.
- **Compatibility:** The system should be compatible with different web browsers, operating systems, and devices to ensure a seamless user experience across a variety of platforms. It should be responsive and adaptable to different screen sizes and resolutions.
- **Maintainability:** The system should be built using modular and well-structured code that is easy to maintain and update. It should adhere to coding best practices and standards, allowing for future enhancements and modifications.
- **Compliance:** The system should adhere to applicable laws, regulations, and industry standards, such as data protection and privacy laws, consumer rights, and e-commerce regulations. It should ensure compliance with relevant policies and guidelines to protect customer rights and maintain ethical practices.

2.4 Feasibility Study

We will evaluate the advantages and disadvantages of our method with the aid of such a feasibility analysis. We can use the information throughout this feasibility report as well as its considerations like a stable platform for evaluating where and when to move ahead. Furthermore, it will aid in making choices regarding the most appropriate software and hardware combinations:

i. Technical Feasibility:

The technical feasibility analysis evaluates whether the necessary infrastructure and technology are available to support the development and maintenance of the site. It examines factors such as hardware, software, and networking capabilities, as well as scalability and compatibility with various devices and platforms. Since **e-Liquor** is an e-commerce site for alcohol delivery, the required infrastructure and technology are readily available within the market. Additionally, it is compatible with all major browsers and devices to ensure a seamless user experience across various platforms.

ii. Economic Feasibility:

There is minimal expenses involved with this e-commerce website. The development process is straightforward and easy to understand. As a result, there is no significant investment required for testing or training. Our website can continuously evolve by adding new features for both buyers and sellers. No additional hardware is necessary for the system, as it can be run efficiently on standard server infrastructure. The recipients of the service (customers) do not need any extra equipment to access or use the platform. This project is economically viable because its potential value and benefits outweigh the initial costs involved in development and maintenance.

iii. Operational Feasibility:

There is no need for extensive instructions to use the product due to its user-friendly design. It can be effectively used in a development environment with minimal effort. Since it addresses the challenges associated with traditional liquor-selling methods and aims to provide a dedicated e-commerce platform for the sale of alcoholic beverages, this approach is operationally viable.

iv. Schedule Feasibility:

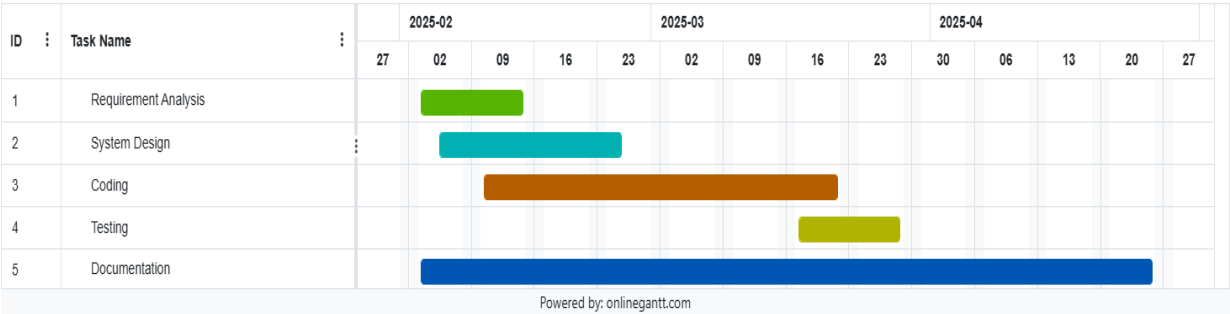


Figure 3 Gantt Chart [one after another]

CHAPTER 3: SYSTEM DESIGN

3.1. System Architecture:

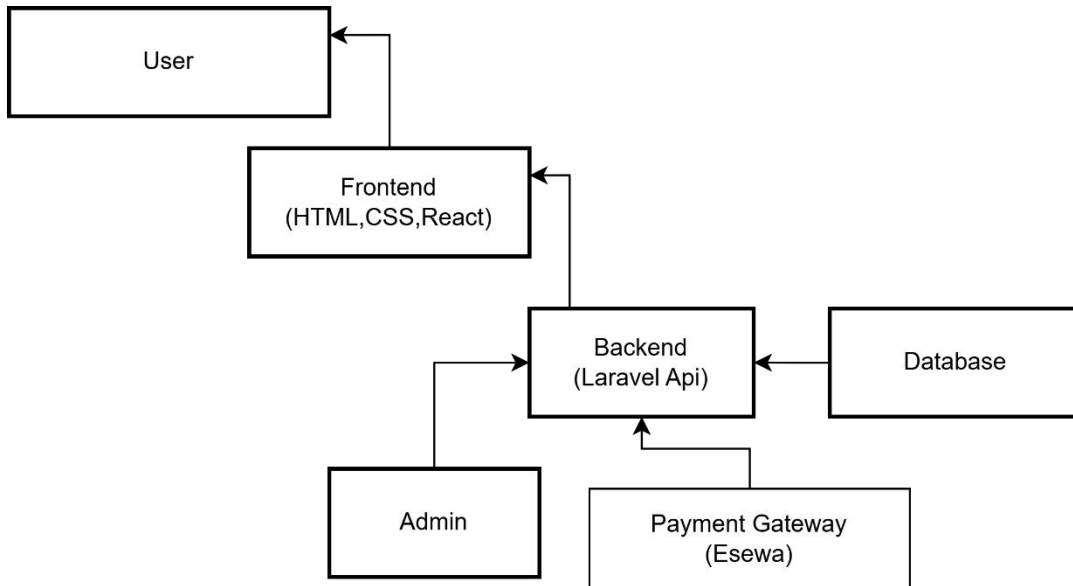


Figure 4 Block Diagram of System

Working Procedure

- **User Interaction:** Customers visit the e-Liquor website to browse and purchase liquor. They interact with the React frontend, which presents a responsive and user-friendly interface. The platform offers easy navigation through product categories, secure payment gateways, and efficient delivery tracking, ensuring a smooth and hassle-free shopping experience.
- **Frontend (React):** Handles UI rendering, product display, form submissions, and validations. Sends HTTP requests (via Axios or Fetch) to the Laravel backend API for actions like registration, login, adding to cart, and placing orders.
- **Backend (Laravel):** Acts as the core application logic layer. Handles user authentication, order processing, product management, and data validation. Communicates with the MySQL database to fetch or store data.
- **Database (MySQL):** Stores structured data such as product details, user profiles, order histories, and admin configurations. Ensures data integrity and fast retrieval.

- Admin Panel: Accessible by administrators for managing products, orders, user queries, and inventory. Interacts with the backend to update and monitor platform activities in real-time.

3.2. Database Design

3.2.1 Class Diagram

This class diagram provides an overview of the database that we have designed for our website.

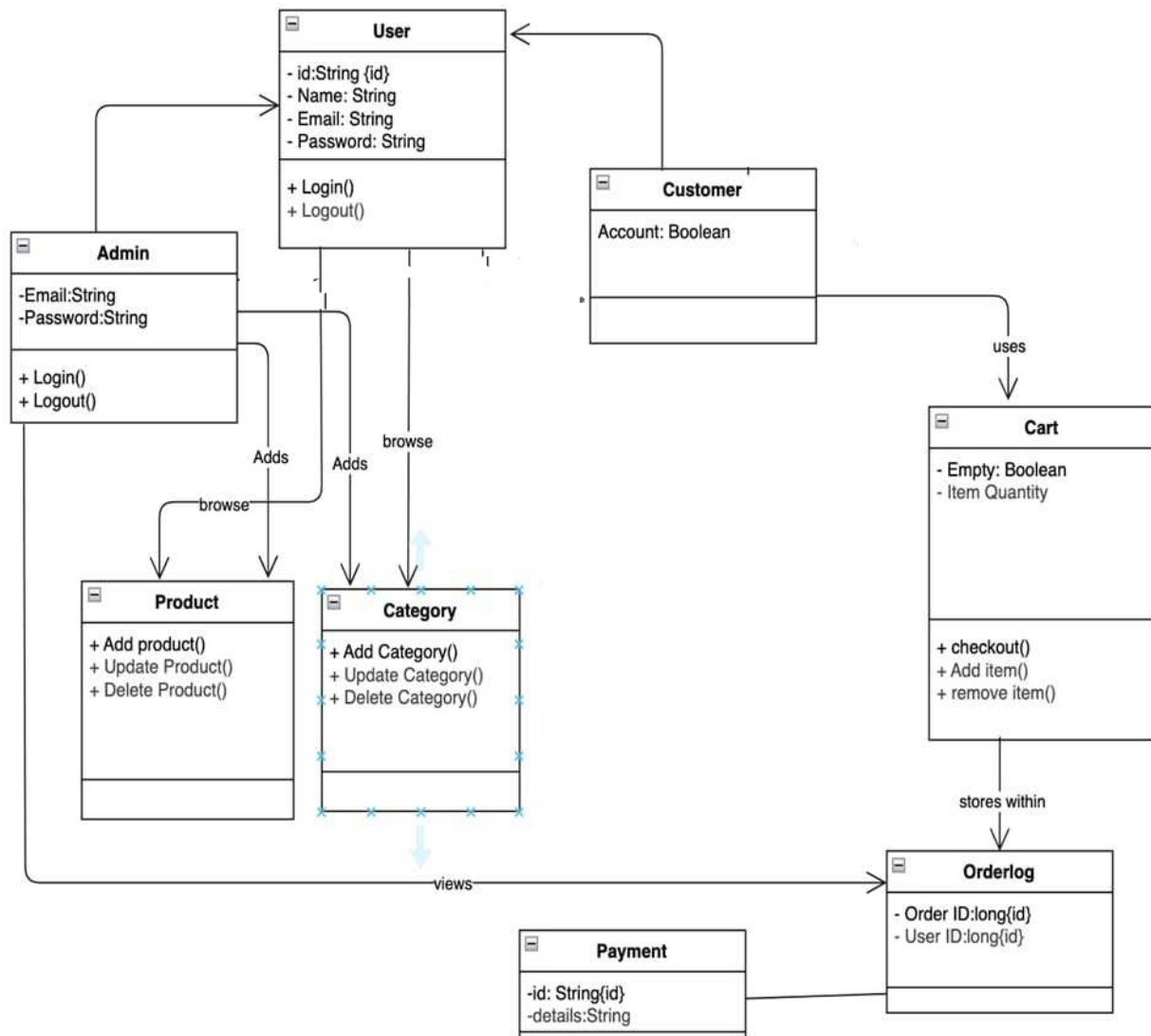


Figure 5 Class diagram for e-Liquor

3.3. Process Design

3.3.1 Sequence Diagram: e-Liquor E-commerce

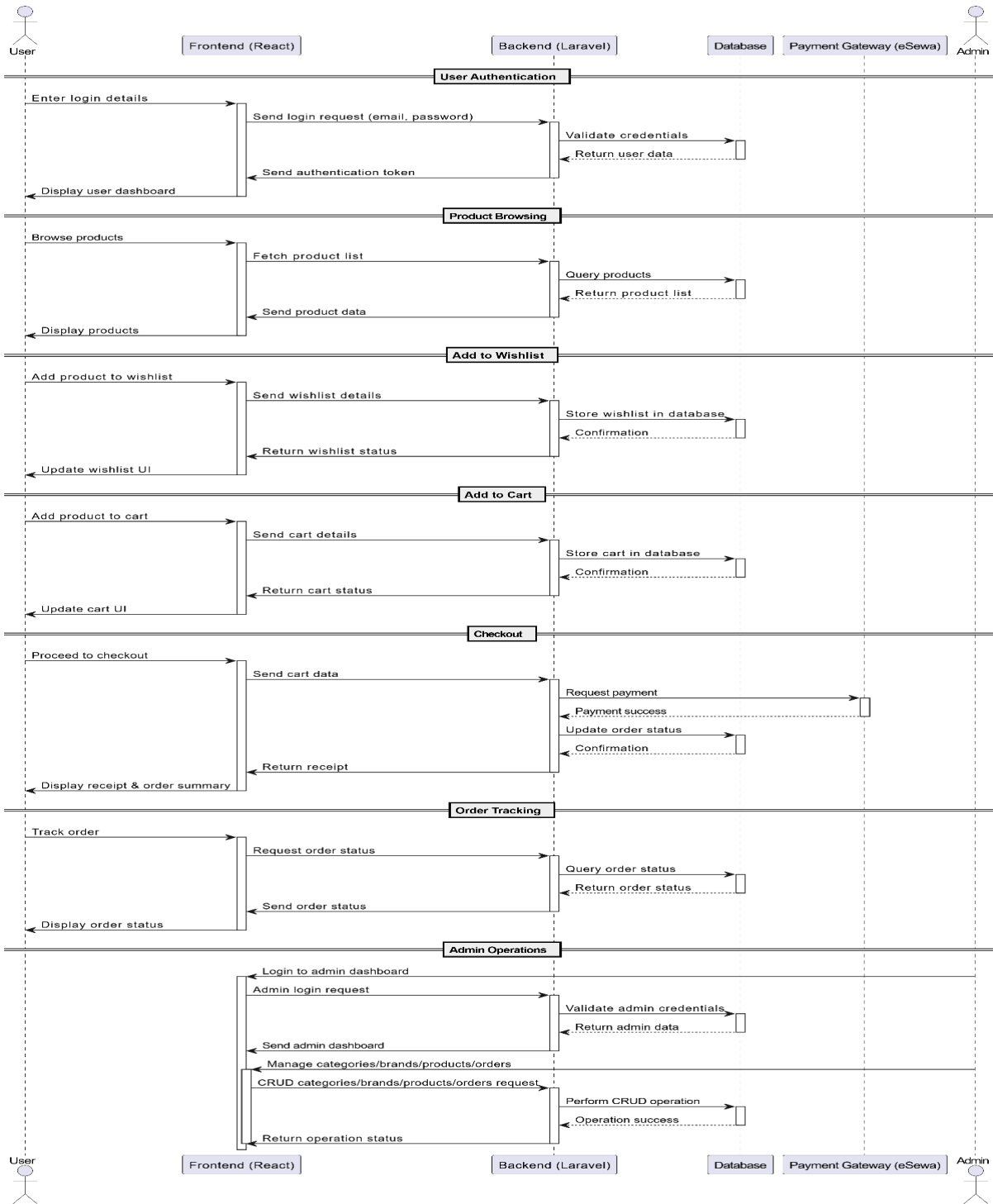


Figure 6 Sequence Diagram for e-Liquor

CHAPTER 4: IMPLEMENTATION

4.1 Implementation

Overview of Implementation

The implementation of the e-Liquor e-commerce platform involved transforming system designs and requirements into a fully functional web application. The development process followed a systematic and structured approach, ensuring that each phase was completed before moving to the next. This method allowed for comprehensive documentation, clear timelines, and a well-defined scope for each development phase, ensuring a smooth development cycle and the delivery of a reliable platform..

- **Development Methodology Adopted**

The Waterfall development model was adopted for this project. It is a linear and sequential approach where progress flows in one direction—through the stages of requirement analysis, system design, implementation, testing, deployment, and maintenance. Each phase must be completed before the next beginning, ensuring a disciplined and well-documented development cycle.

4.1.1 Tools Used

- **Laravel (PHP):** Laravel is the primary backend framework used in this project. It is a robust PHP framework that simplifies the development of web applications. Laravel provides built-in tools for routing, authentication, database management, and more, making it ideal for handling the backend logic of e-Liquor.
- **MySQL:** MySQL is the database chosen for this project. It is a reliable and high-performance relational database management system that integrates seamlessly with the Laravel backend. It is used to manage and store data, including user profiles, product listings, and transaction records.
- **HTML/CSS/JavaScript:** HTML and CSS are used for structuring and styling web pages. With Bootstrap, a responsive design is ensured, making the platform accessible and user-friendly across devices. JavaScript is utilized for client-side interactivity, such as form validation and dynamic content updates.

- **React:** React is used to build the frontend of the application. Its component-based architecture and efficient rendering help create a dynamic and responsive user interface, providing an optimal experience for customers browsing and purchasing eyewear products.
- **Draw.io:** Draw.io is employed to create various diagrams such as flowcharts, use case diagrams, sequence diagrams, and more. These diagrams help in visualizing the system's architecture and the flow of data, ensuring better clarity during development

4.2 Testing

4.2.1 Unit Testing

We employed an instance of input along its related outputs to conduct unit testing. Unit tests separate a piece of code as well as ensure that it is valid.

Table 4. 1 Test case for User Registration

S. N	Test id	Test description	Expected Result	Actual Result	Remarks
1	TC-01	Register a new customer by entering name, email address, password and confirm password	Redirect to the homepage with the current user auth.	Redirected to the homepage with the current user auth.	Pass
2	TC-02	Login with the registered email and password	successfully logged in after activating their account	Successfully login after receiving their account	Pass
3	TC-03	Enter valid admin email address	Display dashboard for the admin user	Displayed dashboard for the admin user	Pass
4	TC-04	Enter incorrect details to register.	The account should not be registered	Registration Failed	Pass

Table 4. 2 Test Case for User Login

S. N.	Test id	Test Detail	Expected Result	Actual result	Remarks
1	TC- 01	Enter valid email address and password	Homepage with products are displayed.	Displayed Homepage with products.	Pass
2	TC- 02	Enter invalid email address and password	Error message displayed.	Displayed error message.	Pass

Table 4. 3 Test Case for adding new product

S. N.	Test id	Test Detail	Expected Result	Actual result	Remarks
1	TC-01	Select the add product section for adding the new product	Successfully submitted and display in the products section	Displayed in the products section	Pass
2	TC-02	Leaves a required field blank	failed product validation message is displayed	Displayed product validation message	Pass
3	TC-03	Enter a title description that exceeds maximum character limit	Error message should be displayed.	Displayed error message.	Pass

Table 4. 4 Test Case for Ordering and Transaction testing

S. N .	Test id	Test Detail	Expected Result	Actual result	Remarks
1	TC-01	Enter appropriate details to place the order.	Transaction is successful and user is redirected to a conformation page.	Successful transaction	Pass
2	TC-02	Review order details including the item, quantity, price, and shopping information	Order details should match the information	Matched ordered information	Pass
3	TC-03	Enter invalid details to place the order	Error message should be displayed	Displayed error message	Pass

4.2.2 System Testing

Table 4. 5 System Testing

Test id	Test description	Expected Result	Actual Result	Remarks
TC-01	After registration, login is done with user name and password.	Show the Homepage for the user	Showed the Homepage for the user	Pass

CHAPTER 5: CONCLUSION AND FUTURE ENHANCEMENT

5.1 Conclusion

In conclusion, the completion of this project has led to the successful development and implementation of a specialized e-commerce platform for the sale of alcoholic beverages. Through thorough analysis, effective requirements gathering, and the application of a structured development methodology, we have addressed the challenges faced by the liquor retail industry in the digital age. By utilizing Laravel for the backend and React for the frontend, we have built a robust, scalable, and user-friendly web application that meets the needs of both customers and businesses.

The system's key features, including user registration, product listing, secure payment processing, and order management, have been meticulously designed and tested to ensure functionality, performance, security, and usability. With Laravel, we have ensured a secure and efficient backend, while React has enabled the creation of a dynamic and responsive user interface. These technologies together provide a seamless experience for customers purchasing liquor, while offering businesses a reliable platform to manage and sell their products.

By overcoming the limitations of existing online liquor marketplaces, our platform provides a tailored and convenient shopping experience. It also positions *e-Liquor* as a valuable asset in the rapidly growing online alcohol delivery industry. With the successful implementation of this system, we have laid the foundation for future enhancements and scalability, ensuring the platform can continue to grow and meet the needs of both users and businesses.

5.2 Future Enhancement

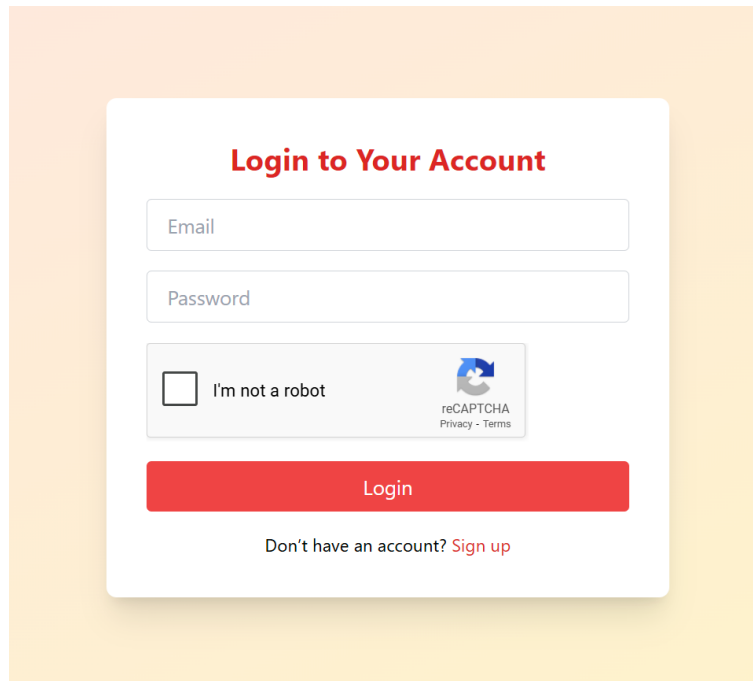
- a) **Enhanced Personalization:** Implement advanced personalization features to cater to individual customer preferences, such as personalized product recommendations and tailored marketing campaigns.
- b) **Integration with social media:** Integrate the e-commerce platform with popular social media channels to expand product reach and leverage the power of social sharing for increased visibility and customer acquisition.

- c) Mobile Application Development: Develop a dedicated mobile application for optimized shopping experiences, leveraging native device features and providing convenience for mobile users.

REFERENCE

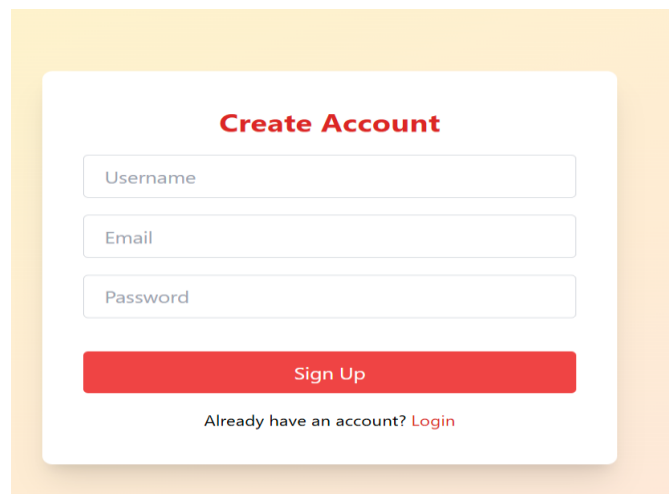
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APPENDIX



The login form is titled "Login to Your Account" in bold red text. It features two input fields for "Email" and "Password". Below these is a reCAPTCHA widget with a checkbox labeled "I'm not a robot" and the reCAPTCHA logo. A red "Login" button is positioned below the reCAPTCHA. At the bottom, there is a link "Don't have an account? Sign up" in red text.

Figure 7: Login Form



The registration form is titled "Create Account" in bold red text. It features three input fields for "Username", "Email", and "Password". Below these is a red "Sign Up" button. At the bottom, there is a link "Already have an account? Login" in red text.

Figure 8: Registration Form

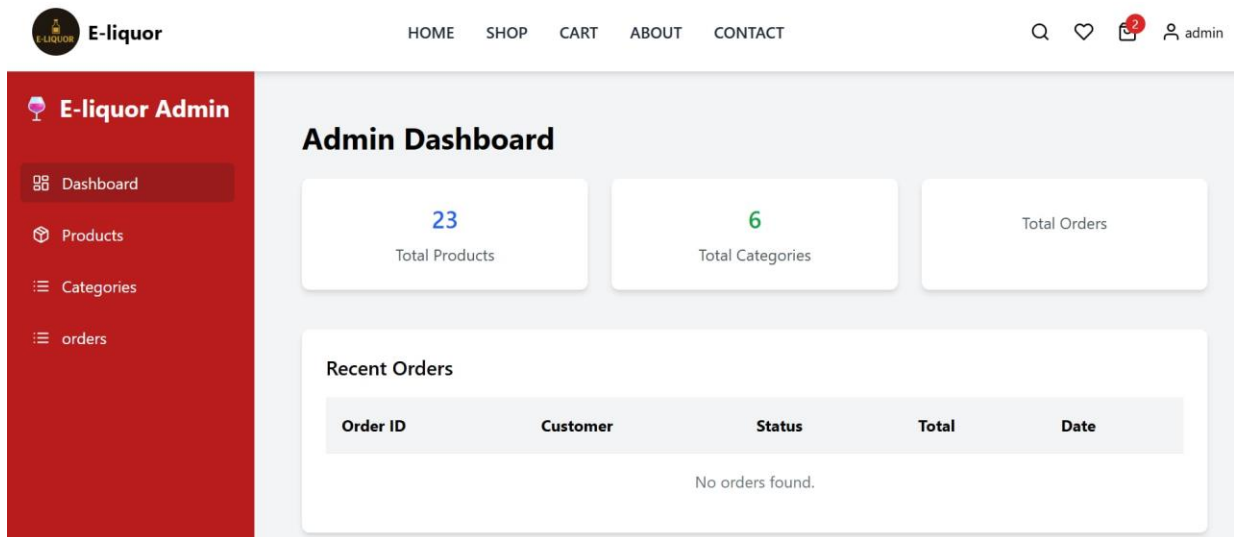


Figure 9: Admin Dashboard

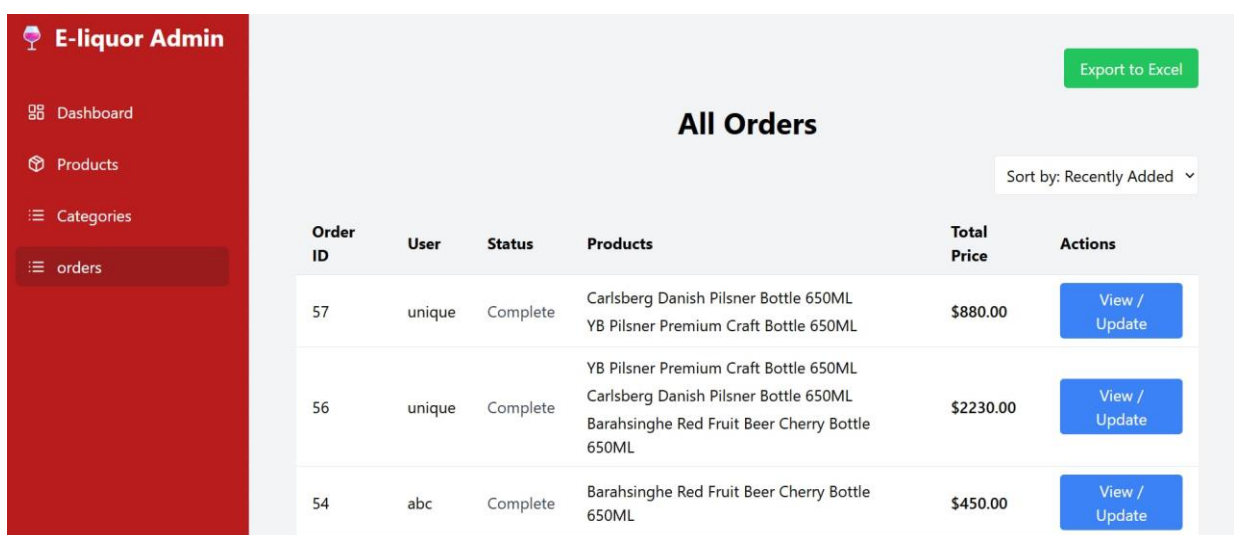


Figure 10: Order Management

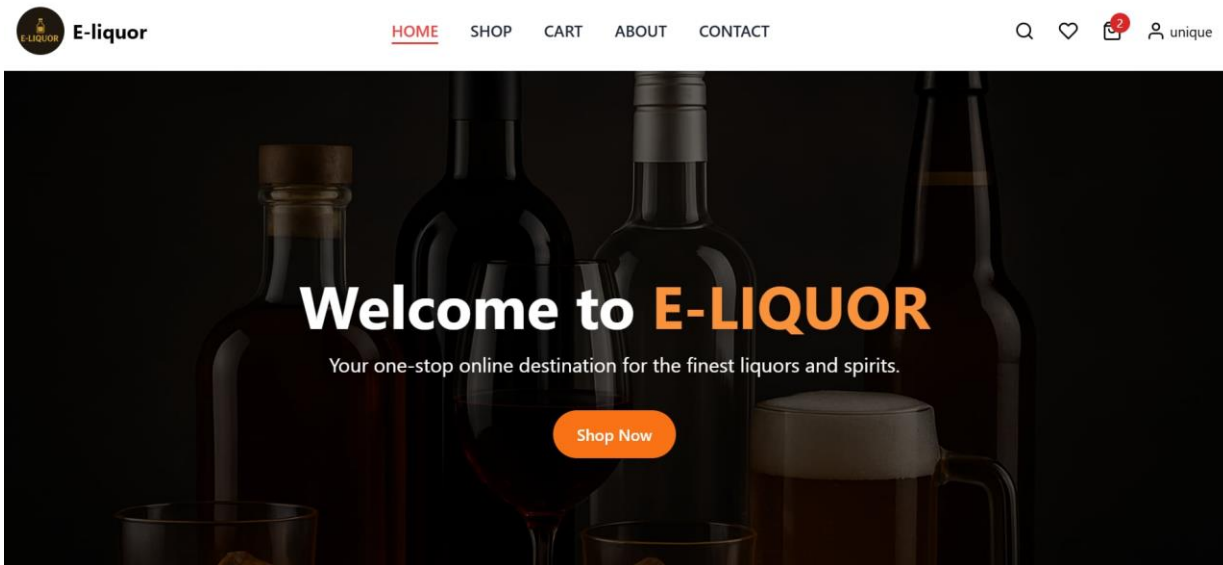


Figure 11:Landing page

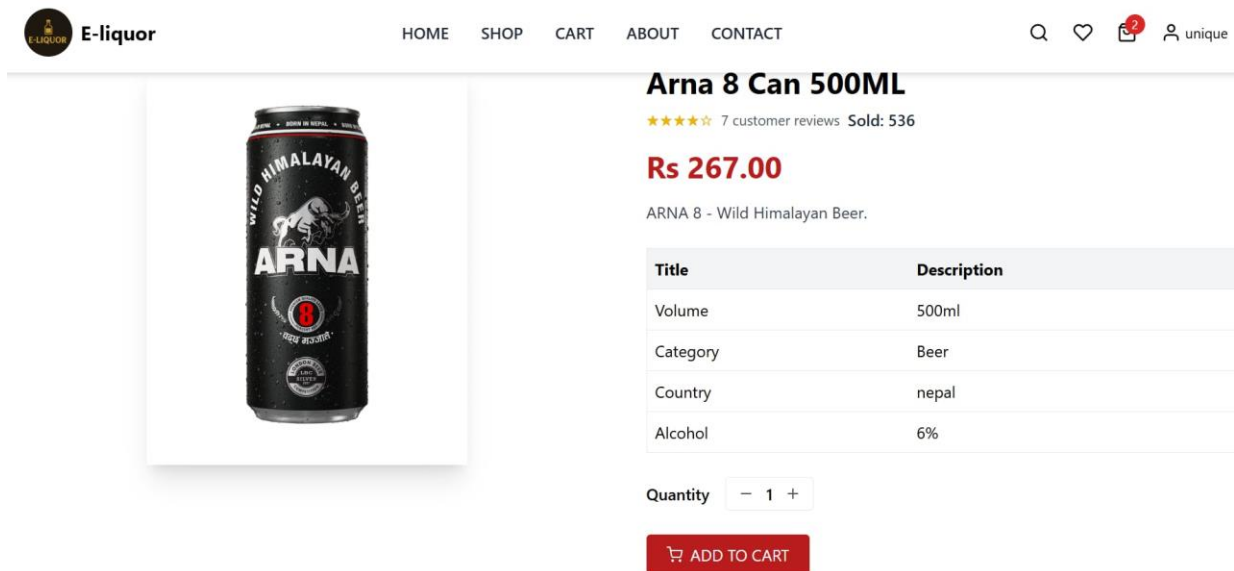


Figure 12:Product Detail Page

Contact Us

Our Shop

We have 3 outlets inside valley, at Jhamsikhel, Imadole and Gwarko

Jhamsikhel Marga, Opposite to Ganesh Mandir

+9779800039393

contact@DaruPatal.com.np

Opening Time

Saturday – Sunday: 6.00 AM – 01.00 AM

Get In Touch

Your email address will not be published. Required fields are marked (*)

Figure 13:Contact Us page

[HOME](#)
[SHOP](#)
[CART](#)
[ABOUT](#)
[CONTACT](#)

Image	Product	Price	Quantity	Total	Remove
	Arna 8 Can 500ML	Rs 267.00	- 1 +	Rs 267	

Cart totals

Number of items	1
Subtotal	Rs 267
Total	Rs 267

Figure 14:Cart page

28

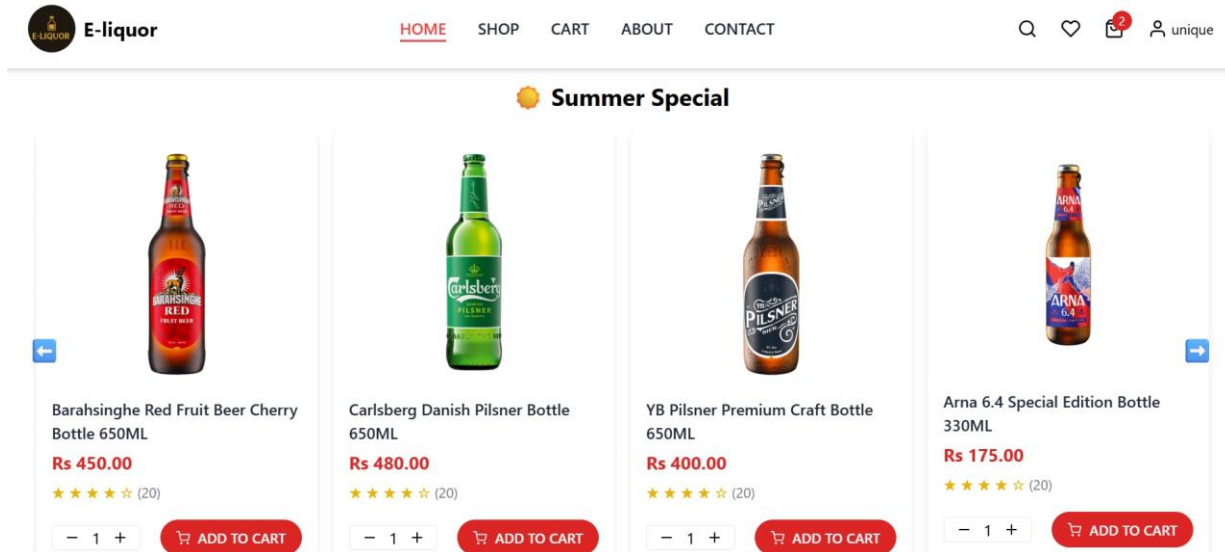


Figure 15:Products Page

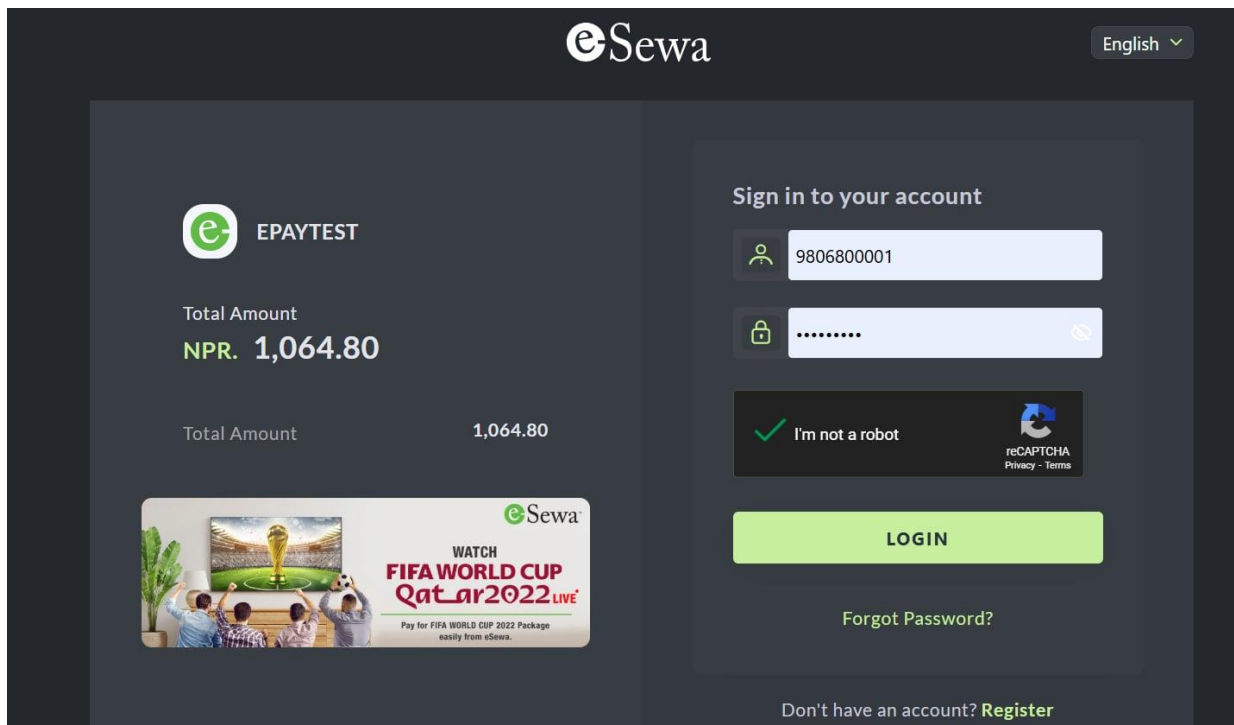


Figure 16:E-sewa Payment page